

Floyd's triangle:

1

2 3

4 5 6

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code for a C program that prints a lower triangular matrix. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function. It uses nested `for` loops to iterate over rows and columns, printing the value of `a` (which increments from 1) in a formatted way. The bottom screenshot shows the program's execution. It prompts the user to "Enter no of rows 3", and the output displays a lower triangular matrix of numbers 1 through 6.

```
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 17 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,a=1;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=r;c++)
{
printf("%3d",a++);
}
printf("\n");
}
getch();
}
```

Enter no of rows 3

```
1
2 3
4 5 6
```

```
TC
Enter no of rows 5
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

```
a=1;
for( r=1; r<=n; r++ )
{
    for( c=1; c<=r; c++ )
    {
        p(a++);
    }
    p("\n");
}
```

$$\frac{n}{3}$$

$$\frac{8}{1}$$

c=1 to r

to 1	_____	1
to 2	_____	2 3
to 3	_____	4 5 6

The image shows two windows of the Turbo C++ (TC) IDE. The top window displays the source code for a C program that prints a triangular pattern of numbers. The code uses nested loops to calculate and print the values. The bottom window shows the program's execution, where the user has entered '3' for the number of rows, and the output is a triangle of numbers: 6, 5 4, and 3 2 1.

```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 55 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,a;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr); a=nr*(nr+1)/2;
for(r=1;r<=nr;r++)
{
for(c=1;c<=r;c++)
{
printf("%3d",a--);
}
printf("\n");
}
getch();
}
```

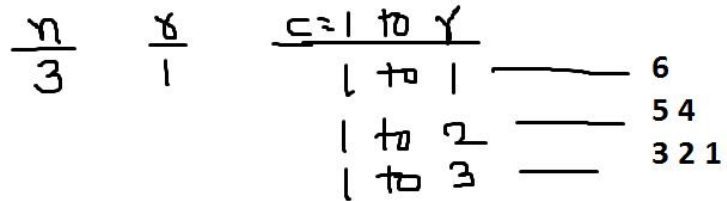
Enter no of rows 3

6
5 4
3 2 1

```

Enter no of rows 5
15
14 13
12 11 10
9 8 7 6
5 4 3 2 1

```



```

for( r=1; r<=n; r++ )
{
for( c=1; c<=r; c++)
{
p(a-- );
}
p("\n");
}

```

$$a = n*(n+1)/2; 3*4/2=6$$

$$a = 5*(6)/2=15$$

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code for a C program that prints a diamond pattern. The code includes headers for `stdio.h` and `conio.h`, and uses `clrscr()` to clear the screen. It prompts the user to enter the number of rows, which is stored in `nr`. Two nested loops are used: the outer loop iterates over rows from 1 to `nr`, and the inner loop iterates over columns from 1 to the current row number. The program uses `printf` to print the pattern, with a conditional statement to alternate between printing `b--` and `a` to create the diamond shape. The bottom screenshot shows the program's execution. The user has entered '4' for the number of rows, and the output displays a diamond pattern of numbers: 1, 3 2, 4 5 6, and 10 9 8 7.

```
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 3 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,a=1,b;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{ b=a+r-1;
for(c=1;c<=r;c++,a++)
{
if(r%2==0)printf("%3d",b--);else printf("%3d",a);
}
printf("\n");
}
getch();
}
```

Enter no of rows 4

```
1
3 2
4 5 6
10 9 8 7
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 17 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,a,b;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{ a=nr-1; b=a+r;
for(c=1;c<=r;c++)
{
if(c==1)printf("%3d",r);else {printf("%3d",b);a--;b=b+a;}
}
printf("\n");
}
getch();
}
```

Enter no of rows 10

```
1
2 11
3 12 20
4 13 21 28
5 14 22 29 35
6 15 23 30 36 41
7 16 24 31 37 42 46
8 17 25 32 38 43 47 50
9 18 26 33 39 44 48 51 53
10 19 27 34 40 45 49 52 54 55
```



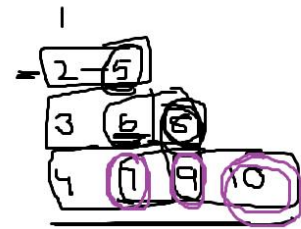
```

for( r=1; r<=n; r++ )
{ a=nr-1; b=a+r;
  for( c=1; c<=r; c++ )
  {
    if(c==1)p(r); else p(b);a--; b=b+a;
  }
  p("\n");
}

```

$\frac{nr}{4}$	$\frac{r}{1}$	$\frac{c=1 \text{ to } r}{1 \text{ to } 1}$
		$1 \text{ to } 2$
		$1 \text{ to } 3$

$a = nr - 1 \implies 4 - 1 = 3 \rightarrow$ 2 1
 $b = a + r \implies 3 + 2 = 5$ ✓
 $3 + 3 = 6$
 $4 + 3 = 7$



$2 + 3 = 5$
 $3 + 3 = 6 + 2 = 8$
 $4 + 3 = 7 + 2 = 9 + 1 = 10$

```

for( r=1; r<=n; r++ )
{ a=nr-1; b=a+r;
  for( c=1; c<=r; c++ )
  {
    if(c==1)p(r); else p(b);a--; b=b+a;
  }
  p("\n");
}

```

$\frac{nr}{4}$	$\frac{r}{1}$	$\frac{c}{1}$
	✓	
	2	1
	3	1
	4	1

$\frac{a=nr-1}{4-1=3}$	$\frac{b=a+r}{3+2=5}$
$\frac{3}{2}$	$3+3=6+2=8$
$\frac{1}{1}$	$4+3=7+2=9$
	$\frac{1}{10}$

	2	5	
-	3	6	8
	4	7	9 10

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 1 Col 33 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=r;c++)
{
if((r+c)%2==0)printf("$ ");else printf("* ");
}
printf("\n");
}
getch();
}
```

```
TC
Enter no of rows 10
$
* $
$ * $
* $ * $
$ * $ * $
* $ * $ * $
$ * $ * $ * $
* $ * $ * $ * $
$ * $ * $ * $ * $
* $ * $ * $ * $ * $
```

\$

1+1=2

*

\$

2+1=3

2+2

\$

*

\$

3+1

3+2

3+3

*

\$

*

\$

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 19 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c; char ch='A';
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=r;c++)
{
if(c==1||r==c||nr==r)printf("* ");else printf("%c ",ch++);
if(ch>'Z') ch='A';_
}
printf("\n");
}
getch();
}
```

Enter no of rows 15

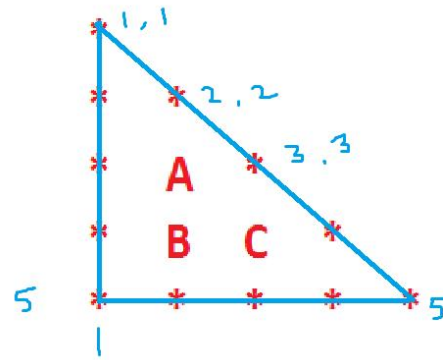
```
*
*
* A *
* B C *
* D E F *
* G H I J *
* K L M N O *
* P Q R S T U *
* V W X Y Z A B *
* C D E F G H I J *
* K L M N O P Q R S *
* T U V W X Y Z A B C *
* D E F G H I J K L M N *
* O P Q R S T U V W X Y Z *
* * * * * * * * * * * *
```

```

ch='A';

for( r=1; r<=n; r++ )
{
for( c=1; c<=r; c++)
{
if(c==1 || c==r || n==r ) p(*); else p(ch++);
}
p("\n");
}

```



```

TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=nr;c++)
{
if(c<=nr-r)printf(" ");else printf("*");
}
printf("\n");
}
getch();
}

```

3:25 PM
16-Sep-24

```
TC
Enter no of rows 4
*
**
***
****
```

```
TC
Enter no of rows 10
*
**
***
****
*****
*****
*****
*****
*****
*****
*****
```

```

for( r=1; r<=nr; r++ )
{
for( c=1; c<=nr; c++)
{
if(c<=nr-1) p(" ");else p(*);
} 1<= 3 - - - *
p("\n"); 2 - - * *
} 3 * * * *

```

$$\begin{array}{r} nr - 1 = 3 \\ 4 - 1 = 3 \\ 4 - 2 = 2 \\ 4 - 3 = 1 \\ 4 - 4 = 0 \end{array}$$

$$\begin{array}{r} c = 1 \text{ to } r \\ 1 \text{ to } 1 = 1 \\ 1 \text{ to } 2 = 2 \\ 1 \text{ to } 3 = 3 \\ 1 \text{ to } 4 = 4 \end{array}$$

```

- - - *
- - * *
- * * *
* * * *

```

```

TC
File Edit Run Compile Project Options Debug Break/watch
Line 1 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=r;c++)printf("*");
printf("\n");
}
getch();
}

```

Monday, 16 September, 2024 3:28 PM 16-Sep-24

```

Enter no of rows 7
*
**
***
****
*****
*****
*****

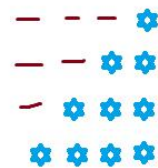
```

```

for( r=1; r<=nr; r++ )
{
    4-3
    4-2
    4-1
    for(s=1;s<=nr-r;s++)p(" ");
    for(c=1;c<=r;c++)p(*);
    p("\n");
}

```

<u>nr</u> - <u>1</u> = <u>5</u>	<u>c=1 to r</u> *
4 - 1 = 3	1 to 1 = 1
4 - 2 = 2	1 to 2 = 2
4 - 3 = 1	1 to 3 = 3
4 - 4 = 0	1 to 4 = 4




```

TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 28 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=r;c++)printf("* ");
printf("\n");
}
getch();
}

```

```

TC
Enter no of rows 10
      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * * *
* * * * * * *
 * * * * * * *
  * * * * * * *
   * * * * * * *
    * * * * * * *
     * * * * * * *
      * * * * * * *

```

```

for( r=1; r<=nr; r++)
{
    4-3
    4-2
    4-1
for(s=1;s<=nr-r;s++)p(" ");
for(c=1;c<=r;c++)p(* );
p("\n");
}

```

<u>nr - 1 = 5</u>	<u>c = 1 to r *</u>
4 - 1 = 3	1 to 1 = 1
4 - 2 = 2	1 to 2 = 2
4 - 3 = 1	1 to 3 = 3
4 - 4 = 0	1 to 4 = 4

```

      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * * *
* * * * * * *
 * * * * * * *
  * * * * * * *
   * * * * * * *
    * * * * * * *
     * * * * * * *
      * * * * * * *

```



```

for( r=1; r<=nr; r++)
{
    4-3
    4-2
    for(s=1;s<=nr-r;s++)p(" ");
    for(c=1;c<=r;c++)p("*");
    p("\n");
}

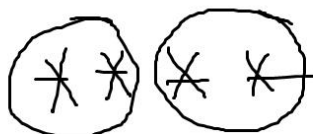
```

$nr - r = 5$ $c = 1 \text{ to } r$ $*$
 $4 - 1 = 3$ $1 \text{ to } 1 = 1$
 $4 - 2 = 2$ $1 \text{ to } 2 = 2$
 $4 - 3 = 1$ $1 \text{ to } 3 = 3$
 $4 - 4 = 0$ $1 \text{ to } 4 = 4$

```

      *
     * *
    * * *
   * * * *
  * * * * *

```



$\frac{8}{1}$

3

$\frac{c}{1 < 2} \quad ***$
 $2 = 2$
 $1 < 3 \quad ****$
 $2 < 3$
 $3 = 3$

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 22 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
while(!kbhit())
{
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)cprintf(" ");
for(c=1;c<=2*r-1;c++){textcolor(random(16));cprintf("*");}
printf("\n");
}
}
```

TC

```
*****
 * * * * *
 * * * * *
 * * * * *
 * * * * *
 *
 * * *
 * * * *
 * * * * *
 * * * * *
 * * * * *
 * * * * *
 * * * * *
 *
 * *
 * * * *
 * * * *
 * * * * *
 * * * * *
 * * * * *
 * * * * *
 * * * * *
 *
*
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
while(!kbhit())
{
textcolor(random(16));
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)cprintf(" ");
for(c=1;c<=2*r-1;c++)cprintf("*");
printf("\n");
}
}
}
```

```
TC
*****
*****
      *
     ***
    *****
   *****
  *****
 *****
*****
*****
      *
     ***
    *****
   *****
  *****
 *****
*****
*****
      *
     ***
    *****
   *****
  *****
 *****
*****
*****
      *
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 20 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
while(!kbhit())
{
textcolor(LIGHTRED);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)cprintf(" ");
for(c=1;c<=2*r-1;c++)cprintf("*");
printf("\n");
}
}
}
```

```
TC
*****
*****
 *
 ***
*****
*****
 *
 ***
*****
*****
 *
 ***
*****
*****
 *
 ***
*****
*****
 *
 ***
*****
*****
 *
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 1 Col 37 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h> #include<conio.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=2*r-1;c++)printf("*");
printf("\n");
}
for(r=nr-1;r>=1;r--)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=2*r-1;c++)printf("*");
printf("\n");
}
getch();
}

Enter no of rows 5
*
***
*****
*****
*****
*****
*****
***
*

```

